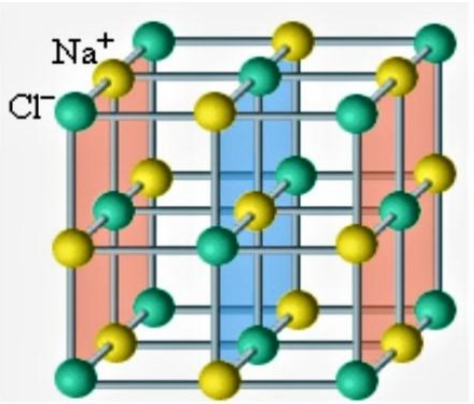
Fizika

9-sinf

Mavzu: **Qattiq jismlarning mexanik xossalari**



Deformatsiya

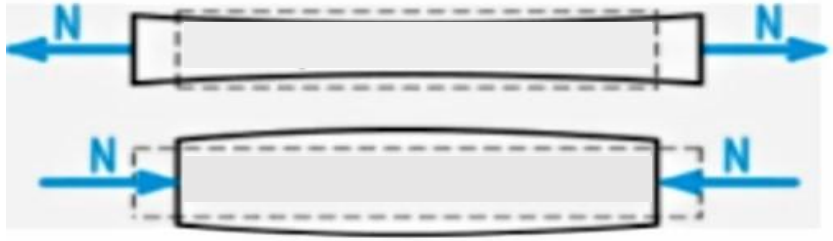
Jismning tashqi kuch ta’sirida o‘z shaklini va o‘lchamlarini o‘zgartirishi ***deformatsiya*** deb ataladi.

Tashqi kuchlarning ta’siri to‘xtatilganidan keyin jism shakli yoki o‘lchami o‘zining dastlabki holatiga qaytsa, bunday deformatsiya ***elastik deformatsiya*** deyiladi.

Jismga qo‘yilgan tashqi ta’sir to‘xtatilganidan so‘ng deformatsiya butunlay yo‘qolmasa, ***plastik deformatsiya*** deyiladi.

Deformatsiya turlari

Cho‘zilish va siqilish

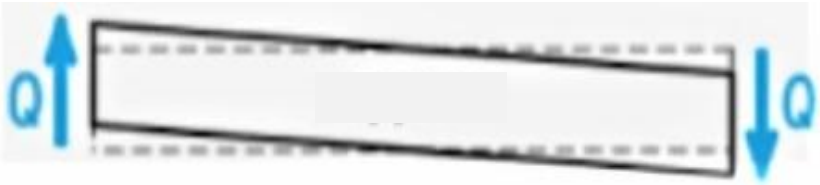




Egilish

Buralish





Siljish

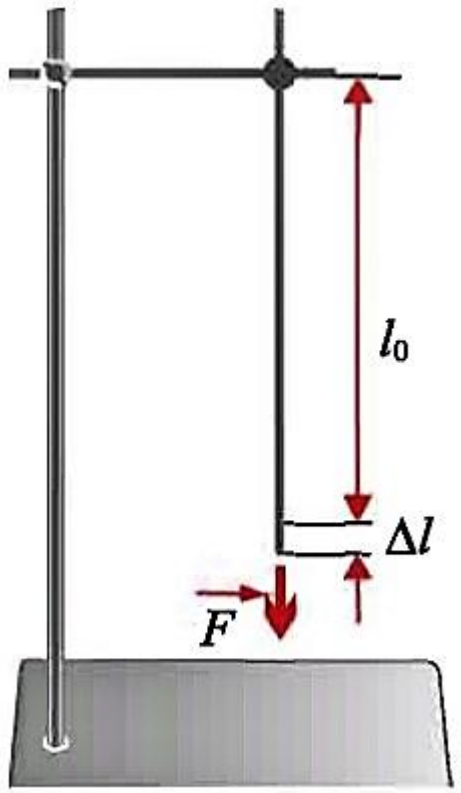
Cho‘zilish deformatsiyasi

$\Delta l -$ absolyut uzayish

$$\Delta l = l - l_0$$

$\varepsilon -$ nisbiy uzayish

$$\varepsilon = \frac{\Delta l}{l_0} \quad \text{yoki} \quad \varepsilon = \frac{\Delta l}{l_0} \cdot 100\%$$



Mexanik kuchlanish

Deformatsiyalangan jismning birlik ko‘ndalang kesim yuziga ta’sir qilayotgan deformatsiyalovchi kuchga son jihatidan teng bo‘lgan fizik kattalik ***mexanik kuchlanish*** deyiladi va u σ harfi bilan belgilanadi.

$$\sigma = \frac{F}{S} \qquad \sigma = E \cdot |\varepsilon|$$

$$\varepsilon = \frac{\Delta l}{l_0} \qquad \frac{F}{S} = \mathrm{E}\frac{|\Delta l|}{l_0}$$

$$F = E \cdot S\frac{|\Delta l|}{l_0}; \qquad \frac{E \cdot S}{l_0} = \mathrm{k}; \qquad F = k \cdot |\Delta l|$$

№	Modda	E, Pa	№	Modda	E, Pa
1	Qo‘rg‘oshin	$1{,}1 \cdot 10^{10}$	4	Mis	$1{,}1 \cdot 10^{11}$
2	Beton	$1{,}6 \cdot 10^{10}$	5	Po‘lat	$1{,}9 \cdot 10^{11}$
3	Alyuminiy	$7 \cdot 10^{10}$	6	Nikel	$2{,}1 \cdot 10^{11}$

Mustahkamlik chegarasi

Materialning mustahkamlik chegarasi modda turiga va uning tayyorlash texnologiyasiga bog‘liq.

	Modda	σ, MPa
1	Beton	48
2	Alyuminiy	50 ÷115
3	Kapron	55 ÷ 80
4	Mramor	100
5	Po‘lat	170 ÷ 700

Elastiklik va mo‘rtlik.

21-mashq 1-masala

Diametri $2\,cm$ bo‘lgan po‘lat arqonga og‘irligi 30 kN bo‘lgan yuk osilgan. Arqondagi mexanik kuchlanishni aniqlang.

Berilgan:

$d = 2\ cm = 0{,}02\ m$

$P = 30\ kN = 30\ 000\ N$

$\pi = 3{,}14$

Formula:

$$\sigma = \frac{F}{S} = \frac{P}{S}$$

$$S = \frac{\pi d^2}{4}$$

$$\sigma = \frac{4P}{\pi d^2}$$

Topish kerak: $\sigma - ?$

Yechish: $\sigma = \frac{4 \cdot 30\ 000\ N}{3{,}14 \cdot (0{,}02\ m)^2} = 95\ 541\ 401\ \text{Pa} \approx 95{,}5\ MPa$

Javob: $\sigma = 95{,}5\ MPa$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(109-bet)
2. 21-mashqning 2-masalasini yechish.